TOPIC NAME : _____

DAY : Farcia - 50

TIME : DATE : / /

DETAILED STUDY OF COVALENT BOND THEORY

TOPIC NAME : _____ DAY : Faria-50

TIME : DATE : / /

□ The d orbitals are in general too large and too high in energy to take part in hybridization with s and p orbitals, and yet it happens, explain.

The d orbitals takes part in hybridization with s and p orbitals in metal complexes where the central metal atom/ion has empty d orbitals available for bonding. Hybridization involving d orbitals can occur when there is a need to accommodate more than four electron pairs around the central atom. So, d orbitals can participate in hybridization to form additional hybrid orbitals, such as sp^3d or sp^3d^2 which allow for the accommodation of more electron pairs.

□ Describe the σ bonding in the following complexes using VBT, mention their hybridization and molecular structure.

(i) High spin $[Co(NH_3)_6]^{3+}$ (ii) $[CoCl_4]^{2-}$ (iii) $[Ni(CN)_4]^{2-}$

[3rd chapter]

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ What are Bonding and Anti-bonding Molecular orbitals ? Write down the bonding and anti-bonding wave functions for the two MOs. What is overlap integral?

Bonding molecular orbitals are those result from the constructive overlap of atomic orbitals, leading to lower energy and increased stability of the molecule. These bonding orbitals are obtained when two atomic orbitals combine together in such way that their lobes with ++ / -- signs.

Antibonding molecular orbitals arise from the destructive overlap of atomic orbitals, processing higher energy and decreased stability. Antibonding MOs are obtained when two atomic orbitals overlap with their lobes +- sign.

If the wave function of an atomic orbital is ψ_A and another atomic orbital is ψ_B,
Then, the wave function of a bonding MO is given by,

$$\psi_b = \psi_A + \psi_B$$

&, the wave function of an antibonding MO is given by,

$$\psi_a = \psi_A - \psi_B$$

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

The overlap integral is a mathematical tool in quantum chemistry that quantifies the degree of overlap between two atomic orbitals in a molecule. It essentially calculates how much the wave functions of these orbitals occupy the same region of space. $\int \psi_A \psi_B d\tau$ is termed as overlap integral.

□ Draw the probability function diagram for bonding ψ_B^2 and Anti-bonding ψ_a^2.

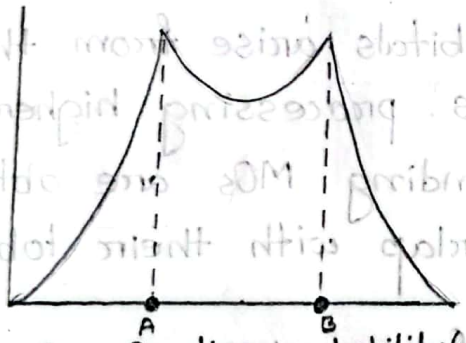


Fig: Bonding probability (ψ_b^2)

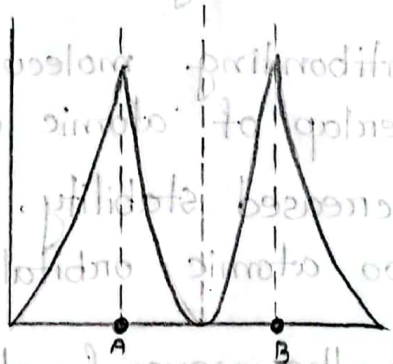


Fig: Anti bonding probability (ψ_a^2)

□ Draw ψ-r graph for both bonding and anti-bonding molecular orbital.

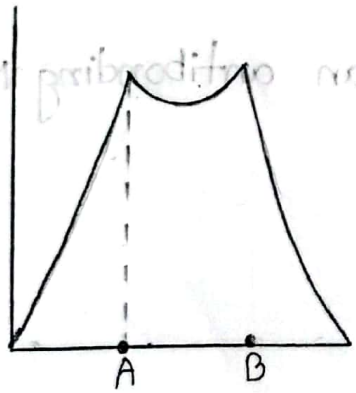


Fig: ψ_b

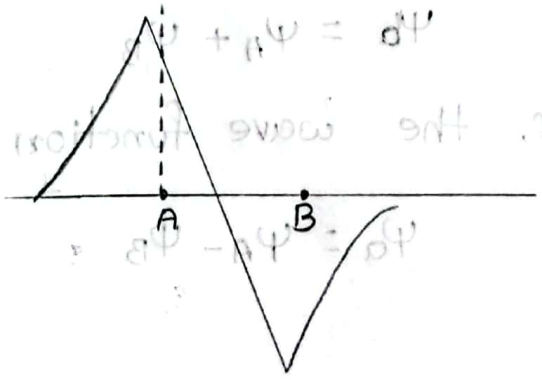


Fig: ψ_a

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ How would you account for the fact that, N_2 is unreactive and diamagnetic but molecular O_2 is paramagnetic?

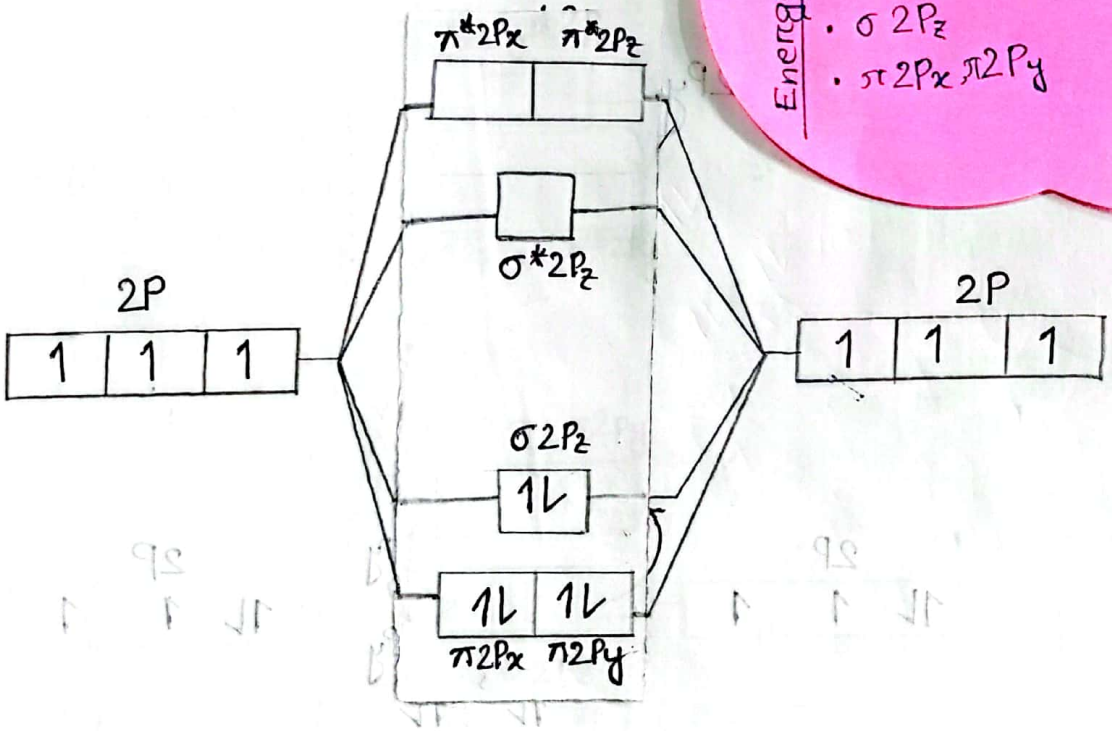
N (7): $1s^2 2s^2 2P_x^1 2P_y^1 2P_z^1$

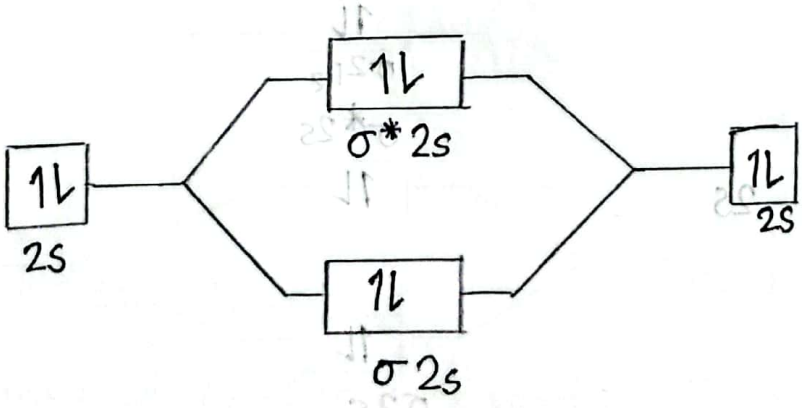
when two N atoms are joined together to form N_2 molecule, they form molecular orbitals. The MO of the out-shell is given bellow:

Atomic number (1-7):

Energy ↑

- $\pi^* 2P_x$, $\pi^* 2P_y$
- $\sigma^* 2P_z$
- $\sigma 2P_z$
- $\pi 2P_x$, $\pi 2P_y$





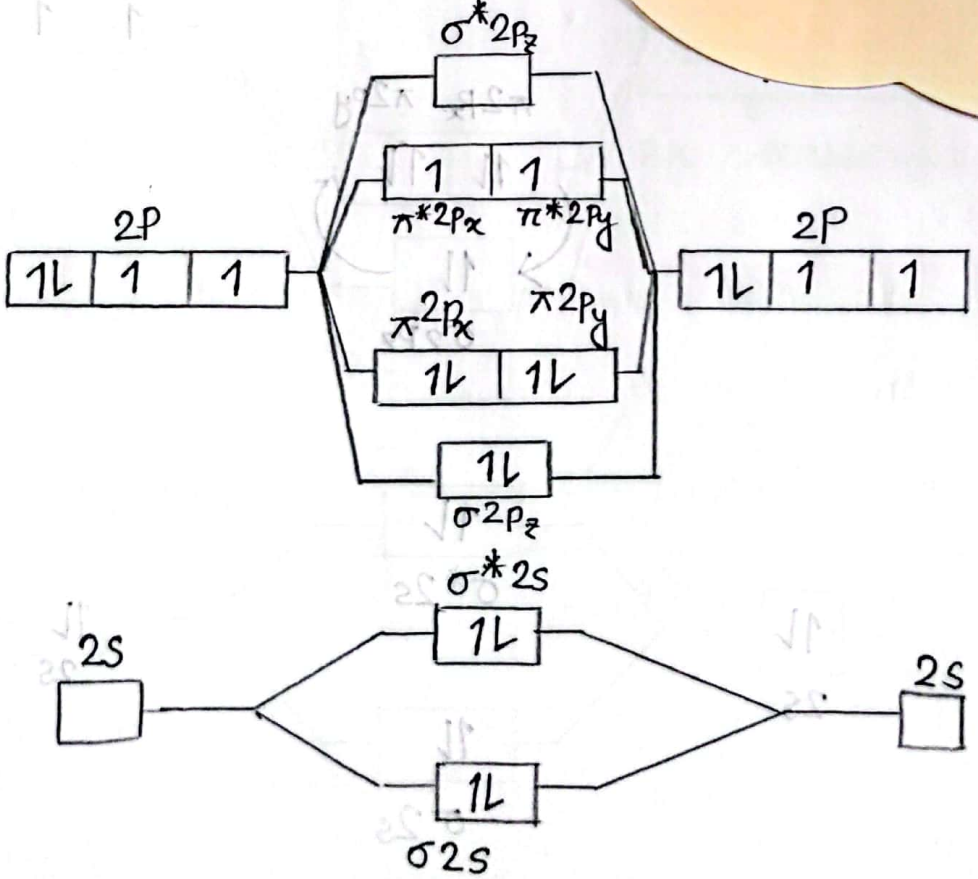
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

From the MOs of N_2 molecule it is seen that all the electrons are paired. Thus nitrogen contain one σ bond and two π bonds. As triple bond contains very high bond enthalpy, it remains unreactive at normal condition with almost every component. Again, as molecular N_2 doesn't contain any unpaired electron, it fails to show any magnetic property. i.e. remains as diamagnetic.

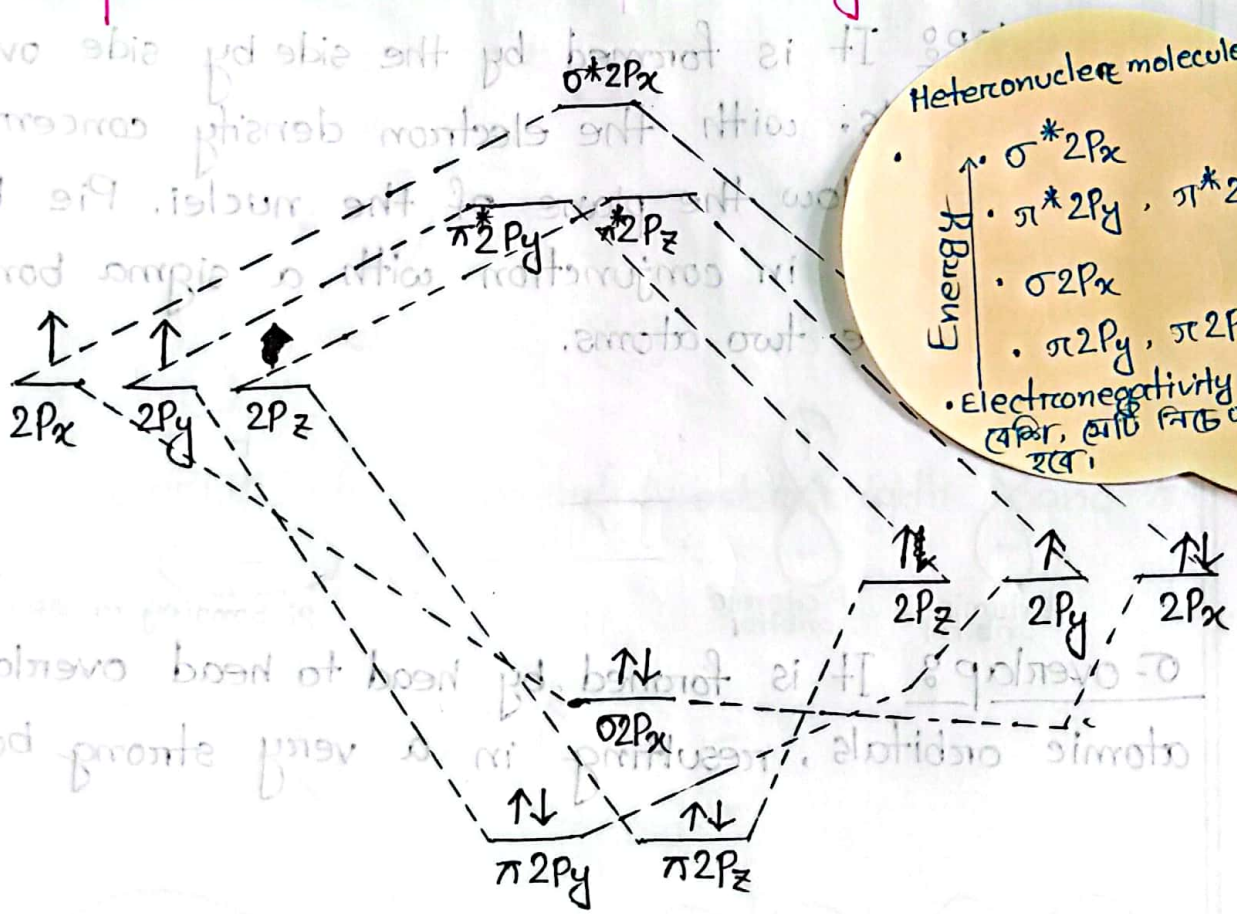
$O(8) : 1s^2 2s^2 2p_x^2 \; 2p_y^1 \; 2p_z^1$

- Atomic no (8 - other)
- Energy ↑
  - $\sigma^* 2p_z$
  - $\pi^* 2p_x$, $\pi^* 2p_y$
  - $\pi 2p_x$, $\pi 2p_y$
  - $\sigma 2p_z$

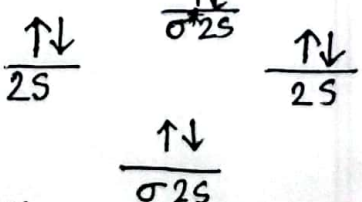


From the energy diagram we can see that, antibonding $\pi^* 2P_x$ and $\pi^* 2P_y$ orbital are singly occupied in accordance to the Hund's rule. Since, there are two unpaired electrons with parallel spin, O_2 acts as paramagnetic molecule.

□ Provide a MO diagram for heterodiatomic molecule CO and compare bond order and bond length in CO and CO+.



C-atom CO - Molecule O- atom



Bond order $= \frac{1}{2}(8-0) = 3$

or, $= \frac{1}{2}(8-2) = 3$, Bond length = 1.128A°

TOPIC NAME : _____ DAY : _____

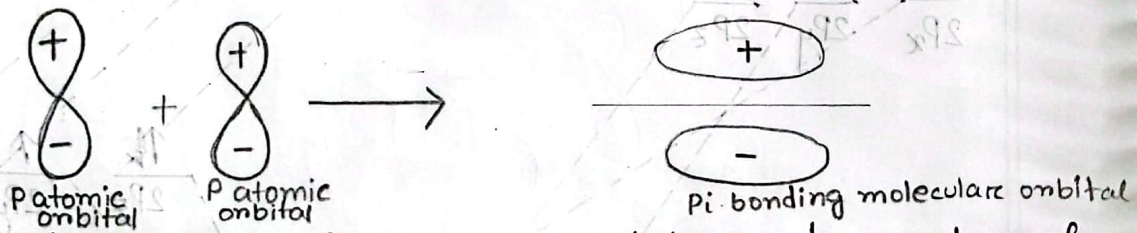
TIME : DATE : / /

In case of CO^+, one electron has been removed from σ^*2s orbital, as it contains the highest energy. Thus bond order = $\frac{1}{2}(8-1) = 3.5$. Bond length is 1.115 Å. As the electron is removed from antibonding orbital, the bond order decreases and bond length decreases than CO.

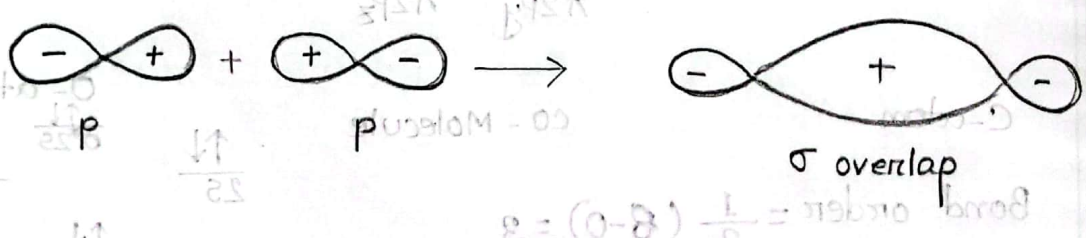
• Bonding orbital $\sigma 2P$, $\pi 2P$, $\pi 2P_z$ এর তুলনায় anti bonding σ^*2s এর শক্তি বেশি।

□ Define π and σ overlap.

π overlap: It is formed by the side by side overlap of p orbitals. with the electron density concentrated above and below the plane of the nuclei. Pie bonds can only exist in conjunction with a sigma bond between same two atoms.



σ-overlap: It is formed by head to head overlap of atomic orbitals, resulting in a very strong bond.



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Draw the π overlap of a metal d orbital with the following:

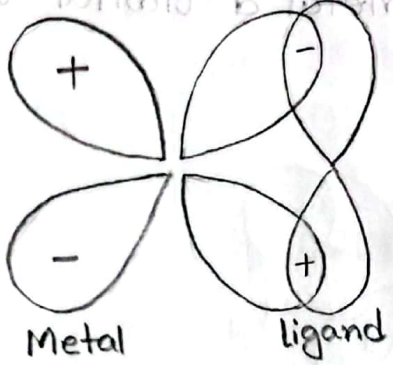
(A) A ligand p orbital

(B) A metal d orbital

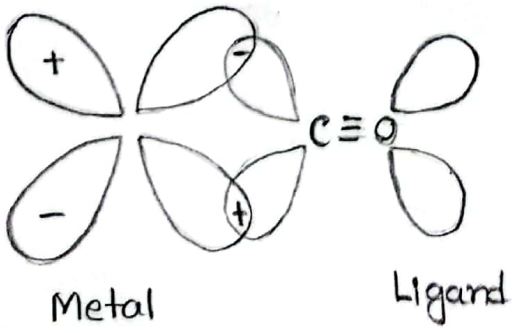
(B) ligand π* orbital

(D) ligand σ* orbital

(A) π overlap of a metal d orbital with a ligand p orbital:



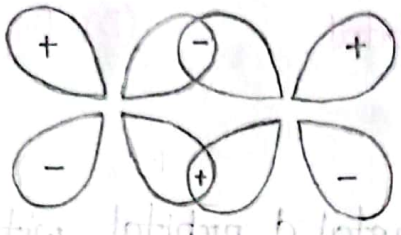
(B) π overlap of a metal d orbital with ligand π* orbital.



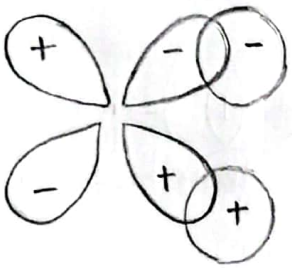
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

(c) π overlap of a metal d orbital with a metal d orbital.



(d) π overlap of a metal d orbital with ligand σ^* orbital.



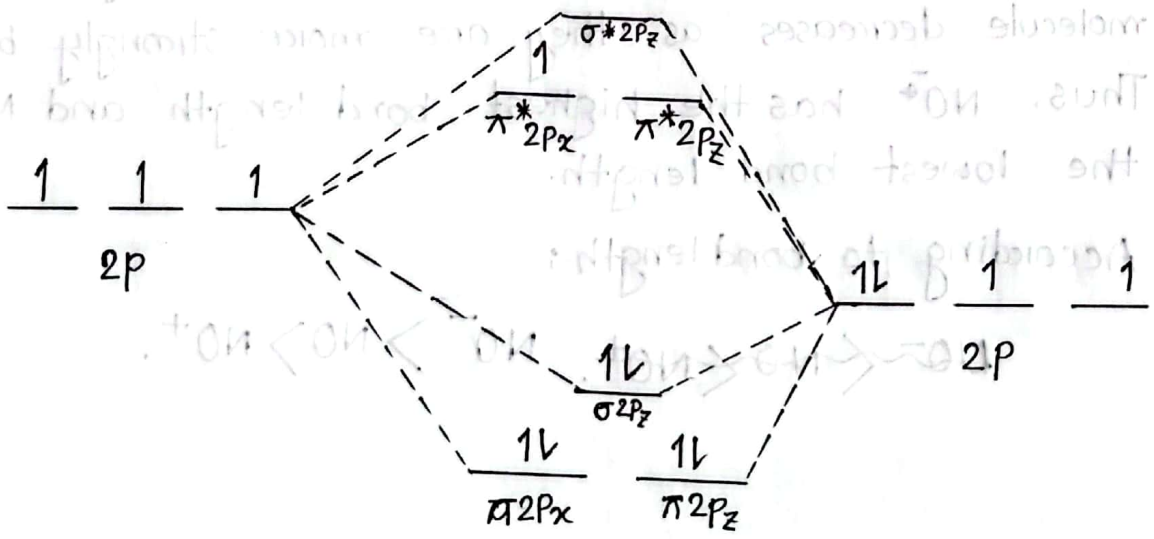
TOPIC NAME : _____ DAY : _____

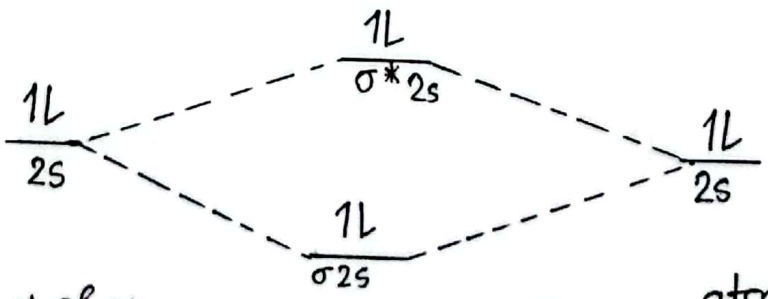
TIME : DATE : / /

□ What are ligand group orbitals (LGO)?

Ligand group orbitals (LGOs) are a concept in molecular orbital theory used to describe the bonding in coordination complexes. They are constructed by taking linear combinations of ligand atomic orbitals that have the same symmetry properties. These LGOs can then interact with the central metal atom's orbitals to form molecular orbitals.

□ Draw the MO diagram of NO molecule. Compare the bond order and the bond length of NO, NO^+, NO^-.





atomic orbital of N

Molecular orbital of NO

atomic orbital of O

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Bond order of NO $= \frac{1}{2}(8-3) = 5/2 = 2.5$

In case of NO^+, one electron has been removed from $\pi^* 2P_x$ orbital.

Thus bond order of $NO^+ = \frac{1}{2}(8-2) = 3$

Again, in case of NO^-, one electron has been added to the $\pi^* 2P_y$ orbital.

Bond orbital of $NO^- = \frac{1}{2}(8-4) = 2$

According to bond order,

~~$NO^- > NO > NO^+$~~. $NO^- < NO < NO^+$

As the bond order increases, the bond length of the molecule decreases as they are more strongly bonded. Thus, NO^- has the highest bond length and NO+ has the lowest bond length.

According to bond length :

~~$NO^- < NO < NO^+$~~. $NO^- > NO > NO^+$.

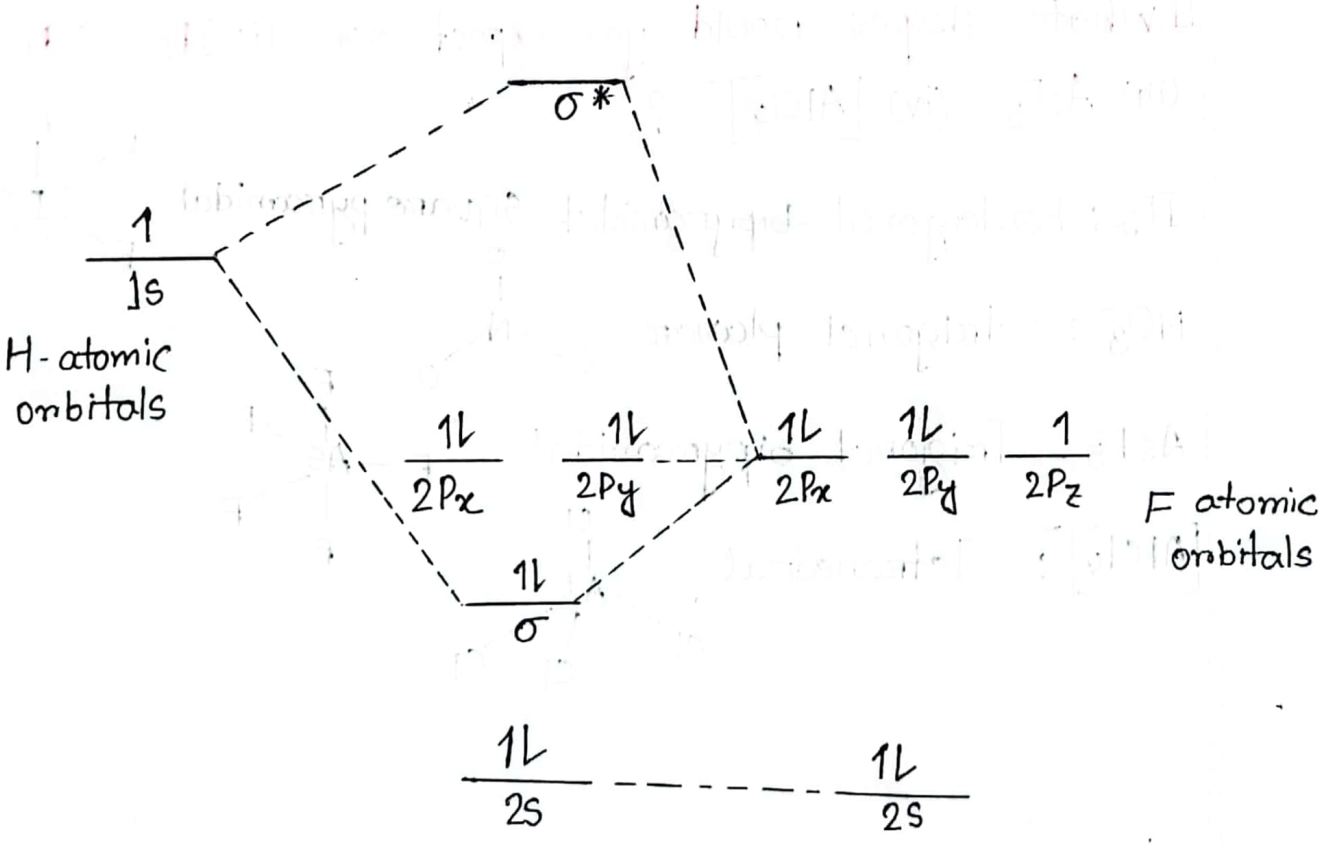
TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Describe the structure of the iodine monchloride, ICl in terms of molecular orbitals.

The MO diagram for ICl shows that the bonding orbital is formed by overlap of a chlorine p orbital and an iodine 5p orbital. There is also a pi-bond component due to sideways overlap of the p orbitals. The lone pairs on iodine occupy non-bonding orbitals. Thus ICl forms a linear structure.

□ Draw the molecular orbital diagram for HF molecule. Clearly sketch how the atomic orbitals interact to form MO.



TOPIC NAME : _____ DAY : _____

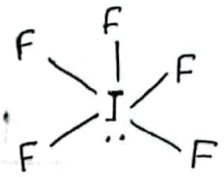
TIME : DATE : / /

1. Hydrogen (H) contributes its 1s orbital.
2. Fluorine (F) contributes its 2s an 2p orbitals but only the 2p orbital interacts significantly with 1s orbital ~~interacts~~ of H due to closer energy levels.

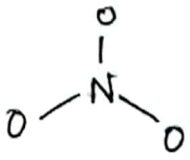
- The overlap of the H 1s and F 2p orbitals leads to the formation of two sigma (σ) molecular orbitals a bonding σ orbital and an antibonding σ^* orbital. The remaining $2P_x$, $2P_y$ orbitals are non bonding because their symmetrical shapes prevent effective overlap with the H 1s orbital.

□ What shapes would you expect for (i) IF_5 (ii) NO_3^- (iii) AsF_5 (iv) $[AlCl_4]^-$?

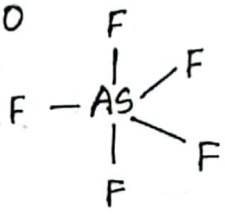
IF_5 : ~~Pentagonal bipyramidal~~ square pyramidal



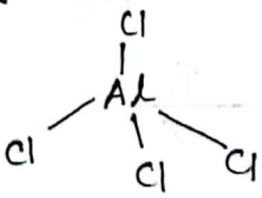
NO_3^- : trigonal planar



AsF_5 : Trigonal bipyramidal



$[AlCl_4]^-$: Tetrahedral



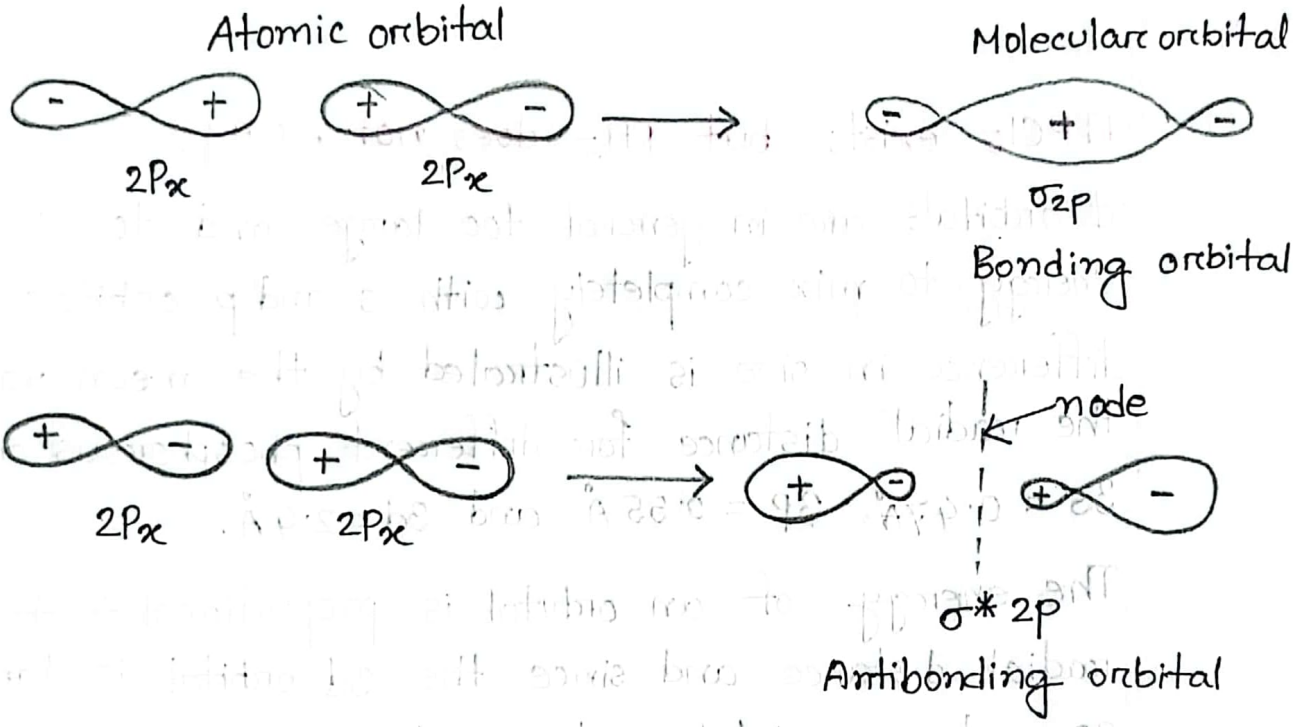
CamScanner

TOPIC NAME : _____ DAY : Farria- SO

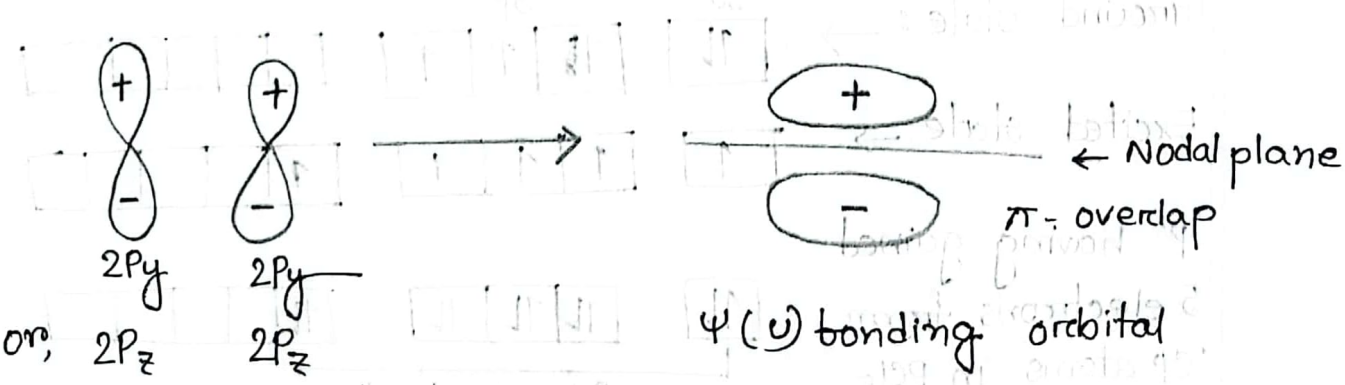
TIME : DATE : / /

□ Draw the bonding and antibonding orbital resulted from 2p orbitals with (i) σ interaction (ii) π interaction.

(i) The bonding and Anti-bonding orbital resulted from 2p orbitals with σ interaction:

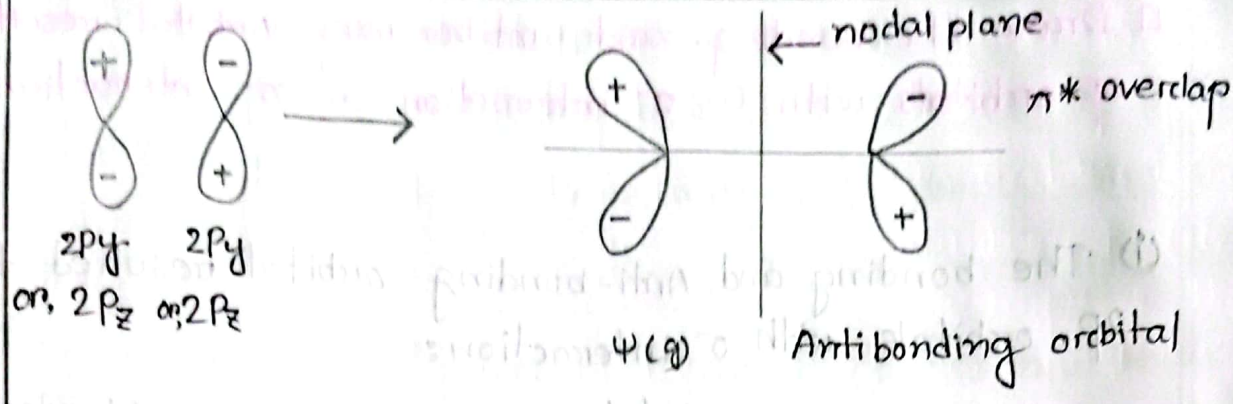


(ii) The bonding and antibonding orbital resulted from 2p orbitals with π interaction:



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /



☐ PCl_5 exists but PH_5 does not, why?

d orbitals are in general too large and too high in energy to mix completely with s and p orbitals. The difference in size is illustrated by the mean values for the radial distance for different phosphorous orbitals: 3s = 0·47Å, 3p = 0·55 Å and 3d = 2·4 Å.

The energy of an orbital is proportional to its mean radial distance and since the 3d orbital is larger than 3p and 3s orbitals, it is also much higher in energy.

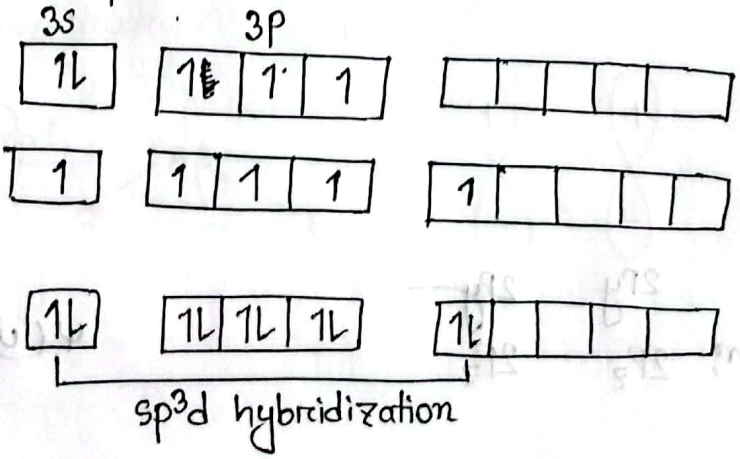
PCl_5

Electronic structure of p atom-

Ground state : →

Excited state →

'P' having gained 5 electrons from 'Cl' atoms in PCl_5



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Chlorine is a highly electronegative atom. In PCl_5. Chlorine atoms pull electron density away from the phosphorus atom, creating a more positive center atom. This fasciliates the promotion of an electron from a lower energy 3s orbital to a higher energy 3d orbital in the P atom. As a result, PCl_5 can achieve a stable electronic configuration with sp^3d hybridization.

In contrast, the weak electronegativity difference between P and H in PH_5 doesn't induce enough promotion. The energy required to promote an electron to a 3d orbital isn't compensated for by the weak bonding interaction with H. Thus, PH_5 doesn't exist.

TOPIC NAME : _____ DAY : Faria-50

TIME : DATE : / /

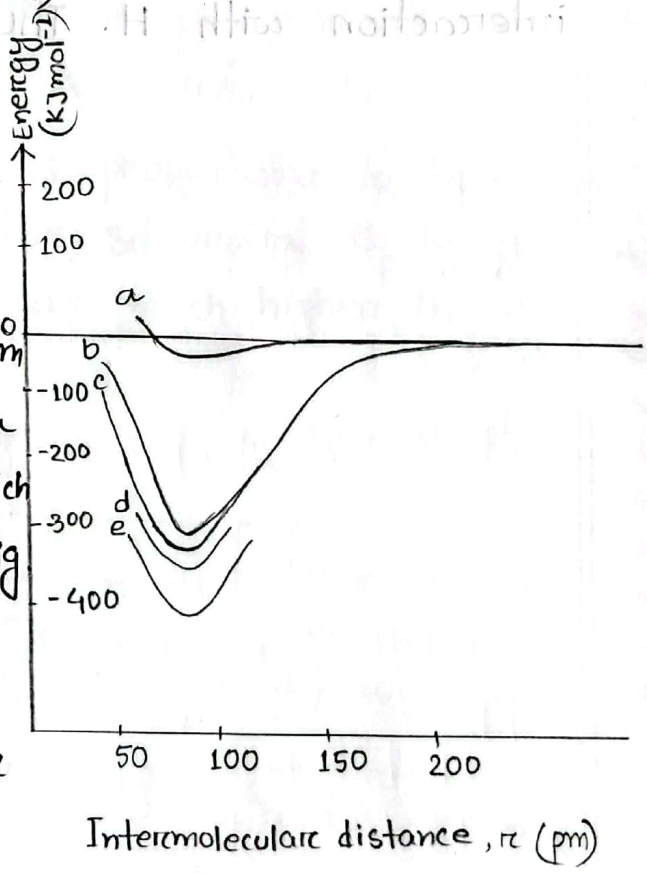
□ In the valance bond theory (VBT) the wave function for a system containing two non interacting hydrogen atoms is given by $\Psi = \Psi_{A(1)}\ \Psi_{B(2)}$ where Ψ_A and Ψ_B designate wave functions of the atoms and the numbers 1 and 2 designate the two electrons. How can you improve the wave functions to account for both covalent and ionic interaction between the atoms.

Ans:

The wave function for a system containing two non interacting hydrogen atoms is given by,

$$\Psi = \Psi_{A(1)}\ \Psi_{B(2)} \text{ ----(i)}$$

But when we solve for energy as a function of distance, we find that the energy curve for equ (i), does indeed have a minimum of about $-24\ kJ\ mol^{-1}$ at a distance of about 90 pm which is defined by curve a in the fig. The actual overved bond distance is 74 pm which is not too different from the approximation but the experimental energy of H_2



is $-458\ KJmol^{-1}$ almost 20 times greater than approximation.

- Again, we are not justified in labeling electrons since all electrons are indistinguishable from each other. So, it would not be sure that electron 1 will always be on atom A and electron 2 on atom B. So the equation can be changed as,

$$\Psi = \psi_{A(1)}\psi_{B(2)} + \psi_{A(2)}\psi_{B(1)} \text{ ---- (ii)}$$

The energy associated with equation (ii), we obtain curve b. The energy is $-303\ KJmol^{-1}$ and the distance has also improved slightly. Thus, the improvement is possible for the allowance of the electrons to exchange.

- A further improvement can be made by calculating effective atomic number Z^* which is less than Z. If we adjust our wave functions, to account for the sheilding from the second electron, we obtain energy curve c-a further improvement. The energy and the distance have been improved upto $-365 KJ/mol$ and 74·3 pm respectively which are very close to the original value.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Although we have allowed the electrons to exchange, we have demanded that they must exchange simultaneously. Obviously this is too restrictive. In some cases, both the electrons can be occupied by an atom and the other becomes vacant, giving raise to the formation of ion pair. It is common to call the arrangement given by equ (ii) the "covalent structure" and to consider the influence of "ionic structures" on the overall wave function:

Now,

$$\Psi = \underbrace{\Psi_{A(1)}\Psi_{B(2)} + \Psi_{A(2)}\Psi_{B(1)}}_{\text{covalent character}} + \underbrace{\lambda\Psi_{A(1)}\Psi_{B(2)} + \lambda\Psi_{A(2)}\Psi_{B(1)}}_{\text{Ionic character.}}$$

Solving the equation, energy has been found -388 KJ/mol and the distance is 74.9 pm which is acceptable. Thus the first approximated equation has been improved.

□ Which of the above mentioned situations leads to greatest stabilization of the system ?

- The wave function where both the ionic and co-valent character is considered leads to greatest stabilization as it includes all possible probabilities. The energy found from the equation (-388 KJ/mol) is closest to the original value (-458 KJ/mol) than other situations and also the distance is closer to the original one.

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Explain resonance in terms of VBT, why the NO bonds in NO_3^- have a bond order of : 1 1/3.

Ans:

Using valance bond theory it is found that more than one acceptable structure can be drawn for a molecule. For H_2 molecule we find both covalent (H-H) and ionic (H^+H^-) structure which lead to the wave function as a linear combination of ionic and covalent functions :

$$\Psi = (1-\lambda)\Psi_{cov} + \lambda\Psi_{ion}$$

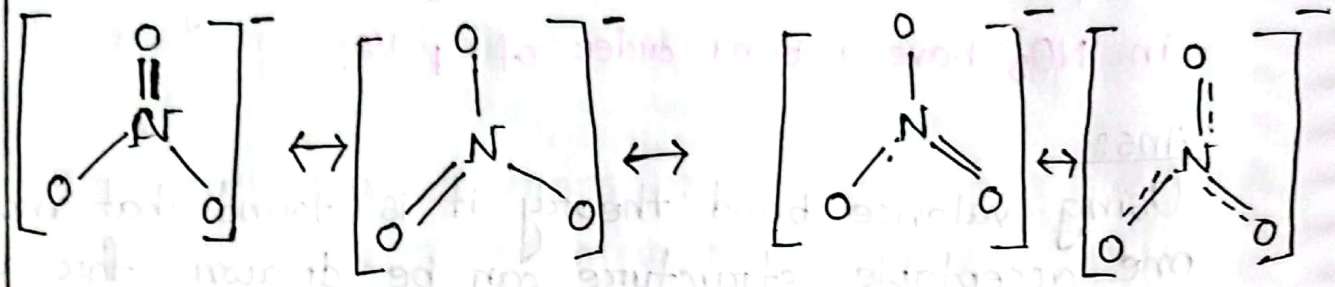
The new function is lower in energy than either of the contributing structures. This is a case of covalent-ionic resonance.

Again, several valid Lewis structures for a single molecule can be drawn, each with different placements of double bonds and formal charges. The actual molecule is considered a resonance hybrid of all the contributing Lewis structure. Thus, the resonance contributes to more stable electronic configuration than single Lewis structure.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Structure of NO_3^- :



Here, 1 orbital of N and three orbitals from three oxygen mix together to form four molecular orbitals.

π_1^* (1 electron)

π_2 (1 electron) $\quad$ π_3 (1 electron)

π_1 (1 electron)

Here, π_1 and π_1^* are bonding and anti bonding orbitals and π_2 and π_3 are non bonding orbitals.

Electrons in bonding orbital = 2, anti-bonding = 0

$\therefore$ Bond order $= \frac{1}{2}(2-0) = 1$

Per bond (N-O) $= \frac{1}{3}$

$\therefore$ overall bond order $= 1 + \frac{1}{3} = 1\ 1/3$

Because of the resonce structure configuration, the bond orderc of NO bonds in NO_3^- has found 1 1/3.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Describe the bonding in $H_2C=C=O$ in terms of valance bond theory.

Ans:

Description of bonding in $H_2C=C=O$:

• Carbon - carbon double bond : The carbon-carbon double bond consists of one sigma bond (σ) and one pie bond (π). The sigma bond arises from the head to head overlap of two sp^2 hybrid orbitals from each carbon. The π bond is formed by the sidewise overlap of unhybridized p orbitals on each carbon atom. These p orbitals are perpendicular to the plane formed by the σ bond and the three sp^2 hybrid orbitals on each carbon.

• Carbon - Hydrogen bonds : Each carbon atom forms single bonds with two hydrogen atoms. These bonds involve the overlap of a sp^2 hybrid orbital from the C atom with a 1s orbital from each H. These bonds are pure σ bonds.

• Carbon - oxygen double bond : Similar to C=C bond, the C=O double bond also has a σ and a π bond. The σ bond results from the overlap of an sp^2 hybrid orbital from C atom with a 2p orbital on the oxygen atom. The pie bond is formed by sideways overlap of a p orbital on the C atom with another p orbital on the O atom.

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ "The idea of delocalization of electrons and resonance have origin in VBT" - Explain.

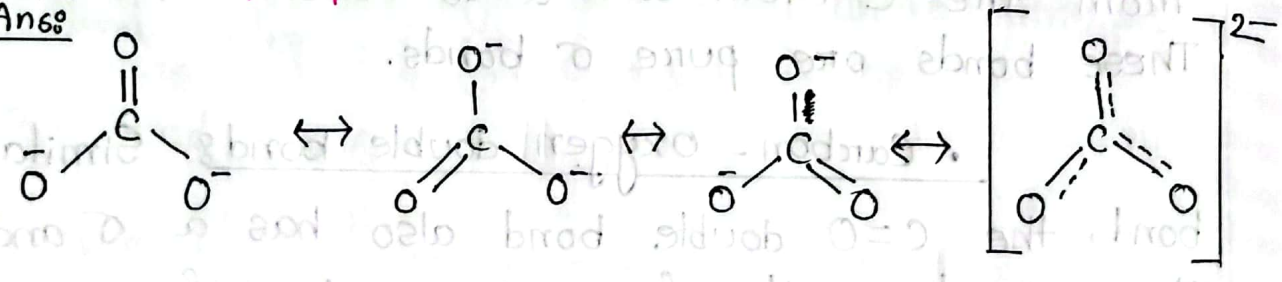
Ans:

The concept of delocalization of electrons and resonce originates from the idea of VBT that in certain molecules, the actual electron distribution can not be accurately represented by a single Lewis structure. This is because the electrons are not localized in specific bonds. Thus resonance occurs when there are multiple valid Lewis structures, each representing a different arrangement of electrons. These resonance structures are not seperate entities but rather contribute to the electron distribution in the molecules originated from VBT.

□ The carbonate ion CO_3^{2-} has a symmetrical trigonal planare structure.

*(i) Draw appropriate resonance structures.

Ans:



TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

(ii) Find out the bond order of C-O bonds.

[** same as N-O bond **]

Ans:

Electrons in bonding orbital = 2

" " anti " " = 0

Bond order $= \frac{1}{2}(2-0) = 1$

per bond (C-O) $= 1/3$

$\therefore$ overall bond order of C-O bonds $= 1 + 1/3$

$= 1\ 1/3$

$\frac{1\downarrow}{\pi_1^*}$ (Anti-bonding)

$\frac{1\downarrow}{\pi_2}$ $\frac{1\downarrow}{\pi_3}$ (Non-bonding)

$\frac{1\downarrow}{\pi_1}$ (Bonding)

(iii) Comment on the hybridization of orbitals on the carbon atom.

Ans:

In CO_3^{2-}, one s orbital and two p orbitals on the carbon atom combine to form three equivalent sp^2 hybride orbitals.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ For B_2 molecule :

(i) Draw the molecular orbital energy level diagram.

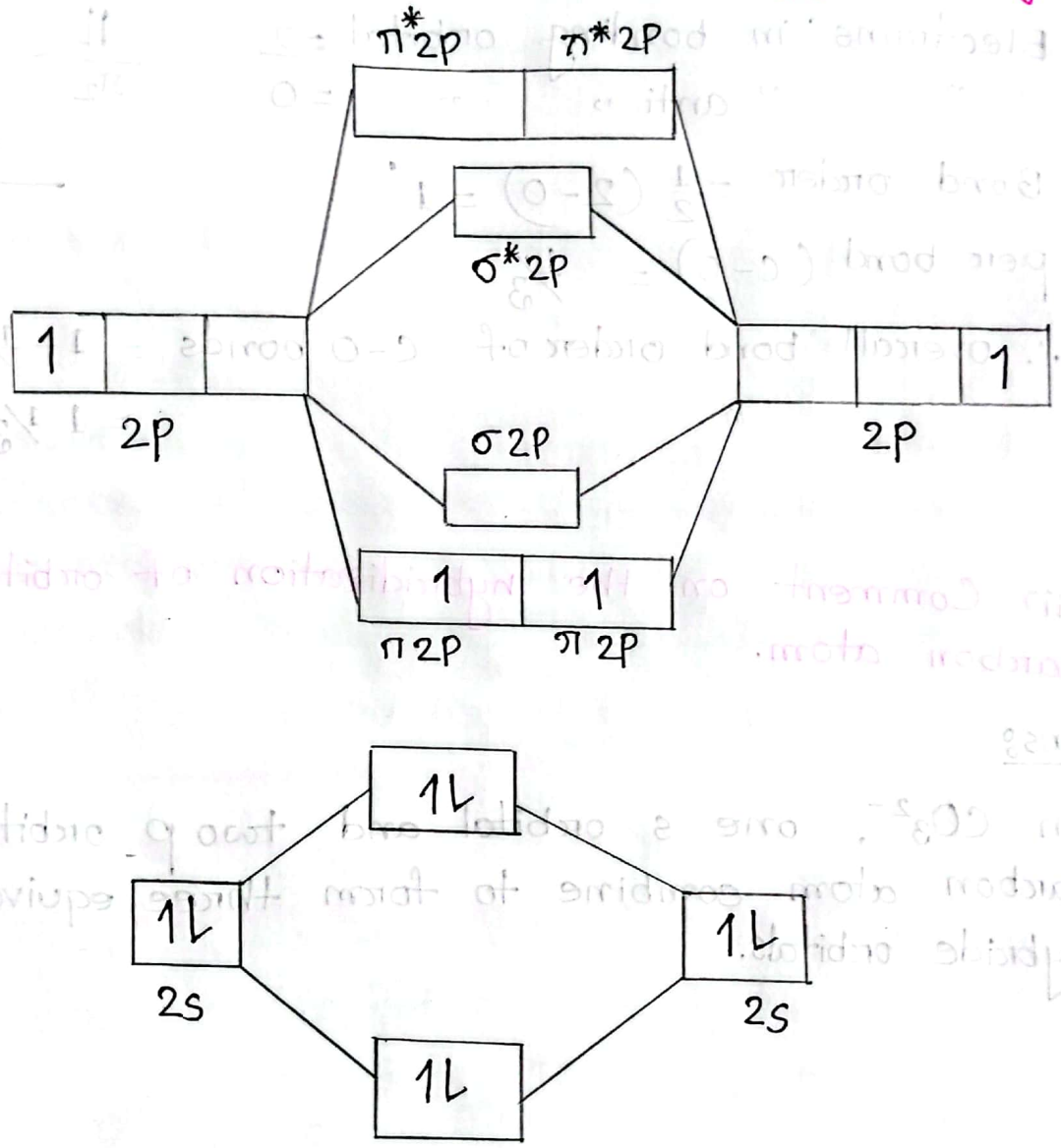


Fig: B_2 molecule.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

(ii) Write the molecular energy configuration.

The molecular energy configuration of B_2 is:

$(\sigma 1s)^2 \ (\sigma^* 1s)^2 \ (\sigma 2s)^2 \ (\sigma^* 2s)^2 \ (\pi 2p_x)^2 \ (\pi 2p_y)^2$.

(iii) Give the bond order.

Electrons in bonding orbital = 4

" in anti " " = 2

$\therefore$ Bond order = $\frac{1}{2}(4-2) = \frac{2}{2} = 1$

(iv) Comment on magnetic property.

As, B_2 has two unpaired electron in it's configuration, it shows magnetism and act as a paramagnetic substance

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

TOPIC NAME : _____ DAY : TIME :

• Size of hybrid orbital is largerr than individual atomic orbital & electron density remains at one side. Thus it leads to better overlap with incoming orbitals.

□ Define hybridization. Why is the use... are preferable over pure atomic orbi... bond ?

Ans:

Hybridization is the mixing of atomic orbitals with the same energy levels giving raise to the same number of new orbitals with equal energy.
Hybrid orbitals are preferred as the followings:

• Better overlap & bond strength: Hybrid orbitals are designed to maximize orbital overlap with other atomic or hybrid orbitals, thus promoting stronger and more stable covalent bonds. This overlap increases the probability of finding electrons between the nuclei, leading to a more stable bond.

• Accomodation of multiple bonds: Hybridization allows atoms to form multiple bonds (double or triple bonds) by mixing different types of orbitals.

• Explanation of molecular geometry: Hybridization helps in predicting and explaining the shapes of molecules. Ex: The tetrahedral shape of methane is explained by sp^3 hybridization of carbon.

• Electronic configuration & stability: Hybrid orbitals distribute electron density more evenly around the nuclei of bonded atoms, contributing to the stability of molecules.

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

• **Compatibility with Molecular Orbital Theory:** Hybrid orbitals are consistent with molecular orbital theory which describes the behaviour of electrons in molecules using linear combinations of atomic orbitals to form molecular orbitals.

□ What do you mean by Molecular Orbital ? How does it differ from Atomic Orbital ?

Ans:

Molecular orbital is a mathematical function describing the wave-like behaviour of an electron in a molecule. Some differences between Atomic and Molecular orbitals are:

Atomic Orbital	Molecular Orbital
1. It is the region having the highest probability of finding an electron in an atom.	1. It is the region having the highest probability of finding electron in a molecule.
2. Formed by electron cloud around the atom.	2. Formed by the fusion of atomic orbitals that have nearly the same energy.
3. Atomic orbitals have fixed shape.	3. Molecular orbitals are shaped according to the type of hybridization.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Explain with suitable example of the formation of MO from AO through LCAO.

Ans:

A classic example of forming Molecular Orbitals from Atomic Orbitals through LCAO method is hydrogen molecule (H_2).

In the formation of H_2 molecule two hydrogen atom combine and each contributes a 1s orbital containing one electron. These have same energy and appropriate symmetry of overlap. Two atomic orbitals combine to form two new molecular orbitals with different energy levels : one with higher energy level and one with lower. The lower energy orbital is termed as bonding MO and the higher energy as antibonding as MO. Electrons fill the bonding MO and antibonding remains vacant. Thus stable H_2 molecule with a single covalent bond is formed

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Class:

□ What is exchange energy?

Ans:

The amount of energy decreased when electrons are exchangable in H_2 molecule rather than being fixed for a particular atom, is known as exchange energy.

□ What are the differences between Valance Bond Theory (VBT) & Molecular Orbital Theory (MOT)?

Ans:

VBT	MOT
1. Valance Band Theory is a basic theory that is used to explain the chemical bonding of atoms in a molecule.	1. Molecular Orbital Theory explains the chemical bonding of a molecule using hypothetical molecular orbitals.
2. Can be applied for diatomic molecules.	2. Can be applied on polyatomic molecules.
3. In this theory, bonds are localized to two atoms.	3. In this theory, bonds are localized to both two atoms and molecules.
4. Resonance plays an important role.	4. Resonance does not play any role.
5. The theory explains about molecules occupying atomic orbitals.	5. The theory explains about the mixing of atomic orbitals to form molecules.
6. According to this theory, oxygen is diamagnetic.	6. Explains paramagnetism of oxygen molecule.

CamScanner

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

7. Uses hybridization to explain geometry of molecules	7. Uses symmetry of atomic orbitals to explain geometry of molecules.
8. Only half filled orbitals in valance shell take part in bonding.	8. All atomic orbitals (filled, half filled, empty) overlap to form molecular orbitals.

□ What are the conditions for orbitals to form MO?

Ans:

Conditions are:

(i) Energy of the participating atomic orbitals should be same or closest enough.

(ii) Their symmetry should be same.

(iii) The number of atomic orbitals are mixed should form equal number of molecular orbitals.

(iv) MO with lowest energy would be bonding, highest energy would be anti-bonding and same energy as the atomic orbital would act like non-bonding molecular orbitals.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Why is He_2^+ / H_2^{2+} more stable than He_2?

Ans:

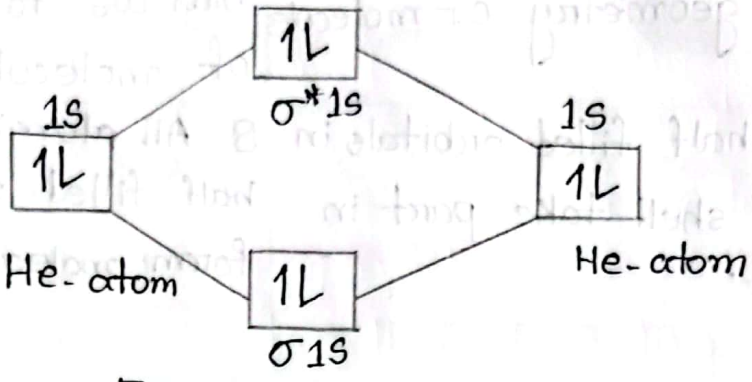


Fig: He_2 molecule.

Bond order of He_2 = ½ (2-2) = 0, As the bond order is 0, He_2 doesn't exist because they can't make any stable bond. But He_2^+ & He_2^{2+},

Bond order of He_2^+ = ½ (2-1) = 0.5

" " of He_2^{2+} = ½ (2-0) = 1, which show that they form bond together. These bonding character leads to greater stability of He_2^+ / He_2^{2+}.

□ Define localized molecular orbital & delocalized molecular orbital.

Ans:

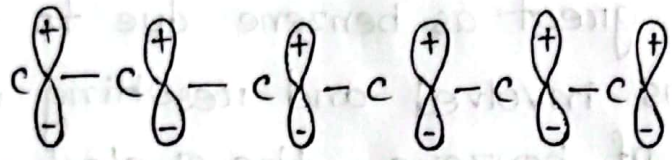
• Localized Molecular Orbital: Localized molecular orbitals are which concentrated in a limited spatial region of a molecule, such as specific bond or lone pair.

• Delocalized Molecular Orbital: Delocalized molecular orbitals are extended over an molecule over more than one specific bond.

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

□ Determine the bond order of C-C bond in benzene.



6 atomic orbitals mix together to form 6 molecular orbitals:
3 bonding & 3 - antibonding orbitals.

Bond order = $\frac{1}{2}(6-0) = 3$ (benzene)

per bond = $\frac{3}{6} = \frac{1}{2}$ & 1 σ bond per C-C bond

↳ π bond

∴ Overall bond order of C-C bond

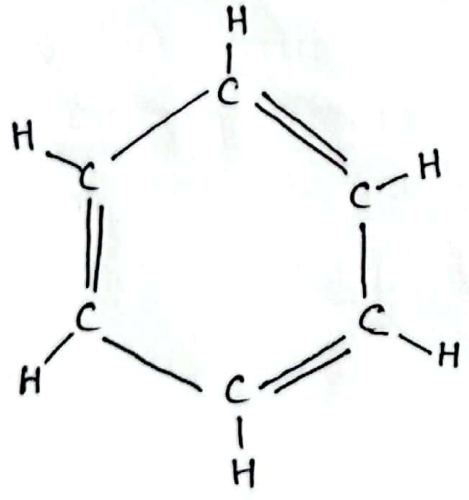
$= 1 + \frac{1}{2} = 1.5$

↳ σ- bond

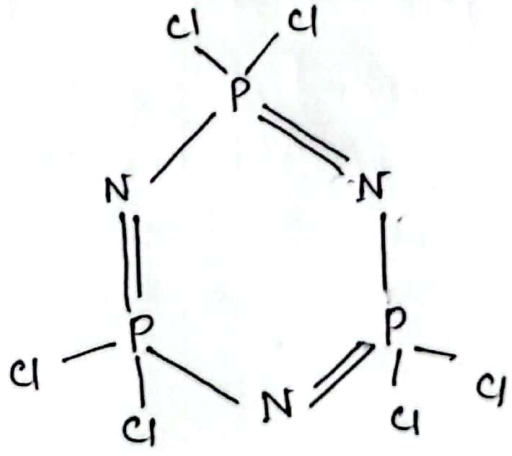
π_1 π_2^* π_3^*

π_2 π_3 π_1

□ Why the delocalization of Phosphonitrile compound is not same as benzene though they have "same" resonance structure ?



Benzene

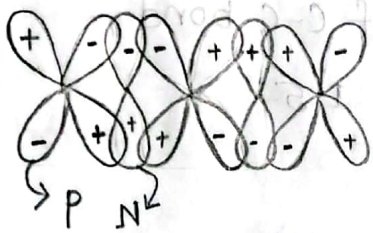


phosphonitrile compound

TOPIC NAME : _____ DAY : _____

TIME : DATE : / /

Though phosphonitrile compound shows some delocalization, that is not as great as benzene due to different nature of atoms involved and resulting electronic effects. In case of benzene, the π electron is delocalized between six carbon atoms which indicates same stability. But in phosphonitrile compound P is larger and less electronegative than N leading to weaker π bonding and reduced delocalization compared to benzene.



The symmetry of first P and last N doesn't match. So, the pie cloud get disrupted leading to poore delocalization.

